

**BIOLOGY 413
PRINCIPLES OF BIOLOGY I**

Start: 8/29/2018 – end: 12/12/2018

Wednesday 1:01pm - 3:50pm

Room: PANDRA P301

Instructor: Dr Megan M Thompson

Office: TBD

Office Hours: TBD and by appointment

Email: megan.thompson@unh.edu**At the end of the semester you should:**

have acquired a basic understanding of biological chemistry, cell structure and function, cell division, genetics, and cellular metabolism.

Required Text: Biology 2e from OpenStax

Print ISBN: 1947172514, Digital ISBN: 1947172522

Available for free from <https://openstax.org/details/books/biology-2e>**Other Materials:**

- Hofmann, A, H. 2016. Writing in the Biological Sciences, 2nd edition. Oxford University Press. ISBN 9780190245603.
- A lab coat is required and may be purchased through the UNH-M bookstore.
- You do not need to purchase a lab manual.

Email Communication:

While I check my email daily, please allow 24 hours for me to reply.

Grades

Item	Points	Requirements
Assignments	5%	In-class activities
Quizzes	10%	13 online quizzes,
Exams	60%	3 exams
Laboratory	25%	

The grade scale is: 94-100%=A; 90-93%=A-; 87-89%=B+; 84-86%=B; 80-83%=B-; 76-79%=C+; 73-75%=C; 70-72%=C-; 66-69%=D+; 63-65%=D; 60-62%=D; Below 59.99%=F.

Assignments: Worksheets or other writing activities will be completed during class and passed in at the end of class. Activities will emphasize content from the current chapter.

Quizzes: The quizzes are available online and located in Canvas/MyCourses under “Modules”. You will be given *three* attempts for the timed quiz and the *best* score will be recorded. Quizzes will be available until **11:59 pm EST on Sunday** on the indicated week. If the quizzes are not completed by that time, then a 0 will be recorded in the grade book. No quizzes are dropped. Allow yourself plenty of time to finish the quiz before the deadline or MyCourses might stop mid-quiz. Each quiz has 10 questions and is timed for 10 minutes. The questions at the end of each chapter are **NOT** assigned.

Exams: The exams will be offered online and are located in MyCourses under “Modules”. During exam days, we will not meet during our designated meeting time. Each exam will cover the content from a particular Module. *No* exams will be dropped and each exam can only be taken *once*. The exams will be offered on the indicated week until **Wednesday at 11:59 pm EST**. Allow yourself plenty of time to finish the exam before the deadline or MyCourses could stop you mid-exam. Incomplete exams will be graded on the work completed. Each exam will have 50 questions and is timed for 60 minutes.

ADA Statement:

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Disability Services Office. The Disability Services Coordinator at UNHM is Jenessa Zurek. Jenessa can be contacted at (603) 641-4383, Jenessa.Zurek@unh.edu, or in person in the Student Services Suite.

Academic Honesty:

The Biology Program at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<https://manchester.unh.edu/current-students/academic-counseling#student-rights>). By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else’s work - in whole or in part - as your own; collaborating on answers for individual assignments, or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report.

Cheating is mainly concerned with copying on exams or in lab, bringing crib notes into an exam or referring to notes or the textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Academic Support/Tutoring:

The Center for Academic Enrichment is a resource for all students enrolled in UNH Manchester courses. Our mission is to help under-prepared students prepare, prepared students advance, and advanced students excel. Trained peer tutors and professional staff provide free assistance with reading and writing in all courses, as well as content in all math and science courses. Support for other content areas is dependent on the background and competencies of current peer tutor staff. All enrolled students are entitled to 1 hour of free one-to-one tutoring per week, per course. In addition, peer and professional tutors may run study groups for you class and provide drop-in math tutoring. Please visit us online at manchester.unh.edu/student/center-academic-enrichment to learn more about our services or to make an appointment with a tutor.

In-Class Time	Topics	Reading	Quiz Due
Aug 29	The Study of Life	Chapter 1	Sept 2
Sept 5	The Chemical Foundation of Life	Chapter 2	Sept 9
Sept 12	Biological Macromolecules	Chapter 3	Sept 16
Sept 19	Cell Structure	Chapter 4	Sept 23
Sept 26	Structure and Function of Plasma Membranes	Chapters 5 and 9	Sept 30
	Cell Communication		
Exam I Due Oct 3 at 11:59 pm EST			
Oct 10	Cell Reproduction	Chapters 10 and 11	Oct 14
	Meiosis and Sexual Reproduction		
Oct 17	Mendel's Experiment and Heredity	Chapters 12 and 13	Oct 21
	Modern Understanding of Inheritance		
Oct 24	DNA Structure and Function	Chapter 14 and 15	Oct 28
	Genes and Proteins		
Oct 31	Gene Expression	Chapter 16	Nov 4
Exam II Due Nov 7 at 11:59 pm EST			
Nov 14	Biotechnology and Genomics	Chapter 17	Nov 18
Nov 21	Metabolism	Chapter 6	Nov 25
Nov 28	Cellular Respiration	Chapter 7	Dec 2
Dec 5	Photosynthesis	Chapter 8	Dec 9
Exam III Due Dec 12 at 11:59 pm EST			

**BIOL 413 Laboratory
Fall 2018****Contact Information:**

Keith Legro

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Email: keith.legro@unh.edu

Office: Rm 550B

Feel free to contact me at any time. Email works best, I will attempt to respond to you as soon as possible.

Lab Schedule:**Module 1: Scientific Method**

Aug 29, 30, 31	Safety, Introduction to Brine Shrimp and Scientific Method
Sep 5, 6, 7	Experimental Design
Sep 12, 13, 14	No Labs
Sep 19, 20, 21	Run Experiment, Data Collection, Statistical Analysis
Sep 26, 27, 28	Communicating in Science; Anatomy of a Journal Article, Conventions, Presentations **Quiz 1 brine shrimp
Oct 3, 4, 5	Student Presentations

Module 2: Cells

Oct 10, 11, 12 Mitosis, Meiosis and Cell Division

Module 3: Molecular Genetics

Oct 17, 18, 19	Crime Scene Evidence Collecting & Blood Typing **Quiz 2 Mitosis/Meiosis
Oct 24, 25, 26	Crime Scene ELISA Test **Quiz 3 Blood Typing
Oct 31, Nov 1, 2	Crime Scene Restriction Fragment Analysis ** Quiz 4 ELISA
Nov 7, 8, 9	Crime Scene Final Analysis & Communicating in Science – Forensic report ** Quiz 5 RFLP
Nov 14, 15, 16	Student Presentations
Nov 21, 22, 23	Thanksgiving No Labs

Module 4: Cellular Respiration and Metabolism

Nov 28, 29, 30	Metabolism
Dec 5, 6, 7	Lab Exam

Grading:

Lab grade will be a total of 125pts made up of the following:

Presentation/Lab Report from Module 1	20pts
Presentation/Lab Report from module 3	20pts
Lab Report from Module 2	10pts
Lab Report from module 4	10pts
Lab Quizzes	25pts
Lab Exam	20pts
Participation/Attendance	20pts
Total	125pts

Participation/Attendance:

Attendance in lab is very important as each lab of a module builds upon another. If you miss a lab you will be responsible for gathering the information needed to come to the next lab ready to perform the lab of that week. Also we will be working as groups during this lab and it is not fair to the others in your group to miss lab as that will leave the other members of the group more work to do. In light of these facts **5 pts will be deducted from your lab grade for each missed lab**. Any more than 2 missed labs will result in possible failure of the lab. Also all within a group must contribute to the assignments in this lab or points may be deducted from those who don't equally participate.

Canvas (My Courses):

All of the materials for this lab are found in your lab section of Canvas. While there may be times where you will be given hard copies of materials you will need for lab there will also be times where you will need to print out and bring the materials to lab. It is also a good way for you to review any videos or interactive programs that we may use in lab.

Electronic Devices:

You are welcome to use electronic devices (cell phones, tablets, laptops) in lab for "lab related" purposes. If you however need, or any other non-lab related use of these devices (i.e. to take or place a phone call) please be respectful of your lab mates and go outside the lab for such activities.

Lab Notebook:

You will need a notebook for lab to record your experimental progress and data in.

Lab Manual:

There is no lab manual that needs to be purchased for this lab.